Analyzing Feature Engineering in ML/DL Models for NIDS Performance

INSE 6180 Project

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***Abstract*—Network intrusion detection systems (NIDSs) are vital in defending against cyber-attacks. This study examines the impact of feature fusion and feature selection on deep learning (DL) models for NIDSs, with the goal of improving their performance. Our approach uses strategic feature engineering to evaluate whether it enhances detection accuracy for the Anomal-E model, which initially lacked significant pre-processing. The analysis explores model performance under various conditions, focusing on the effect of feature fusion and selection on high-dimensional and imbalanced datasets. The goal is to provide insights that can help improve intrusion detection systems in practical applications.**

***Index Terms*—Network Intrusion Detection, Deep Learning, Machine Learning, Feature Engineering, High-dimensional Data, Imbalanced Datasets, Decision Tree, Graph Neural Network, Cybersecurity**

I. Introduction

The evolving digital landscape has increased the complexity of cyber threats, challenging traditional cybersecurity measures. Intrusion Detection Systems (IDSs) are crucial for monitoring and identifying potential threats. Network Intrusion Detection Systems (NIDS), utilizing Machine Learning (ML) and Deep Learning (DL), play a pivotal role in enhancing performance, including capturing elusive zero-day attacks [1], [2].

Despite advancements, applying classical ML techniques (e.g. SVM, DT, KNN) and DL methods (e.g. LSTM, CNN, RNN) in network intrusion detection faces challenges. Dealing with large, high-dimensional datasets can make training and testing supervised ML models time-consuming and costly [3], [4]. Feature selection is crucial for reducing dimensionality and enhancing model efficiency to address these challenges [5], [6].

This study aims to investigate how feature engineering techniques affect the performance of deep learning models in the context of network intrusion detection. Specifically, we focus on the Anomal-E graph neural network (GNN) model and examine how feature selection and fusion can improve its ability to detect anomalies in high-dimensional and imbalanced datasets. Our objectives are: 1) To analyze the impact of feature selection and fusion on deep learning model performance in network intrusion detection, and 2) To determine whether these methodologies can enhance the accuracy and efficiency of the Anomal-E model, ultimately contributing to more robust NIDSs.

II. Background and Literature Review

A. Network Intrusion Detection Datasets

The evolution of network intrusion detection datasets, from the initial KDD Cup'99 to modern datasets like UNSW-NB15, CIC-IDS2017, CSE-CIC-IDS2018, reflects the dynamic landscape of

cyber threats. These newer datasets address limitations and incorporate different attack vectors, show the need for intrusion detection systems to adapt to evolving challenges and showcasing ongoing innovation in dataset development [3], [7]–[10].

B. Machine Learning and Deep Learning in NIDS

ML algorithms like SVM, DT, RF offer robust classification capabilities. DL models, such as auto-encoders (AE) for DDoS detection, restricted Boltzmann machines (RBM), deep belief networks (DBN) for feature extraction, recurrent neural networks (RNN) including LSTM and GRU for sequential data processing, further advance detection methods. These DL techniques, alongside traditional ML approaches, address the evolving landscape of cyber threats with improved accuracy and adaptability [3] [11]. Moreover, GNNs have emerged as a powerful deep learning approach for analyzing non-Euclidean data, such as network traffic represented as graphs, capture complex relationships and inter-dependencies between entities, such as sophisticated network threats for advanced network intrusion detection [12].

C. Importance of Feature Selection and Fusion

Intrusion detection models benefit from effective feature selection. Yin et al. proposed IGRF-RFE, enhancing anomaly detection on UNSW-NB15 [13]. [14] used genetic algorithms with Logistic Regression for feature selection on the same dataset . Janarthanan and Zargari identified an optimal feature subset for UNSW-NB15 [15]. Maajid and Nalina focused on feature reduction using Random Forest for importance scores [16]. Eunice et al. integrated Random Forests into Deep Neural Networks for feature selection [17].

D. Gaps and Opportunities in Current Research

The literature review highlights advancements in intrusion detection, emphasizing the significance of feature selection and feature fusion with differnt models. Our project aims to explore the importance of data pre-processing [4], and provides insights into the workings of machine learning (ML) and deep learning (DL) models in NIDS.

III. Proposed Framework/Model

A. Datasets

In addition to the UNSW-NB15[1] dataset , we will also utilize the CIC-IDS2018[2] dataset, known for its realistic and up-to-date attack scenarios. The UNSW-NB15 contains 42 features, where three instances are non-numeric (categorical) features and 39 are numeric in nature.The UNSW-NB15 contains 10 labels with one of them being benign label. The CIC-IDS2018 contains 42 features and 15 labels with one of them being benign label and others are attack labels. These datasets will help in validating the performance of our proposed models across different network environments.

B. Evaluation Metrics

In network anomaly detection, accuracy, precision, recall, and macro F1-score are crucial metrics for evaluating model effectiveness. While accuracy is commonly used, it can be misleading in imbalanced datasets, as it overlooks the ratio of true positives to false positives or true negatives to false negatives. Precision measures the proportion of true positive predictions among all positive predictions, indicating the rate of false positives. In NIDS, high precision can prevent unnecessary alerts but often results in lower recall, possibly missing some attacks. Recall, or detection rate, is crucial for ensuring most attacks are detected, reducing the risk of missed threats. A tradeoff between precision and recall is common in NIDS, with the ideal balance ensuring a robust system that catches most attacks without generating excessive

[1]UNSW-NB15 Dataset
[2]CIC-IDS2018 Dataset

false alarms. The macro F1-score, balancing precision and recall, provides an overall measure of model performance, even in balanced datasets.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall (Sensitivity)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

C. Feature Selection with Decision Trees

To underscore the importance of feature selection, we adopt a methodology inspired by [4], utilizing the XGBoost algorithm for important feature selection. This process narrows down the feature set to the most informative variables for intrusion detection. We'll assess how improved features can boost the performance of traditional ML models e.g. DT. To show the difference in performance with the feature selection, we compare the performance with and without applying feature selection in the experimentation part.

The UNSW-NB15 dataset was used as training and testing dataset here. The features were normalized using min-max algorithm. The algorithm below, was followed to select the important features:

Algorithm 1 Feature Selection with XGBoost

Require: X: Input features, y: Target variable, threshold: Importance score threshold
Ensure: selected_features: List of selected features
SET selected_features as empty list
TRAIN XGBoost model on (X, y)
for each feature importance score in model **do**
  if importance score $>$ threshold **then**
    ADD feature to selected_features
  end if
end for
return selected_features

D. Feature Fusion with Deep Learning

Our project enhances deep learning models for network intrusion detection using the feature fusion technique (Ayantayo et al., 2023). We employ a Late Fusion model, where subsets of inputs are individually trained and merged. The network comprises three fully connected layers with ReLU activation functions and two dropout layers for improved generalization. In the experiment section we described the performance of the DL model with and without the implementation of feature fusion technique.

For the deep learning model, we employ Adam optimization, a stochastic gradient descent approach, to minimize categorical cross-entropy loss. The loss function, denoted as $E_{CCE}(y^i, y', (x^i, W)) = -\sum_{i=1}^{c} y^i \ln y^i(x^i, W)$ where x^j represents the input examples, y^i denotes class labels,c is the number of classes, and $y^{'}((x^j$, W)) is the predicted output from a Softmax function using the converged weight matrix W associated with the deep learning model. We evaluate this feature fusion strategy using the UNSW-NB15 dataset [8]. The algorithm approached for this technique mentioned here 2.

Algorithm 2 Feature Fusion Algorithm

Require: Pre-processed feature subsets $S_1, S_2, ..., S_n$
Ensure: Classification predictions for input data
for $i \leftarrow 1$ **to** n **do**
  Create a fully connected network for feature subset S_i
  Each layer should have a ReLU activation function
  Include dropout layers to enhance generalization
  Train the fully connected network using feature subset S_i
end for
Merge feature subset representations into one vector.
Feed the fused vector into a classification layer
Include a softmax classifier layer to generate predictions
Output node count should align with the classification problem's class count.
return Classification predictions for input data

E. Graph Neural Network for Network Intrusion Detection

To explore the capabilities of deep learning, particularly for data represented as graphs, we propose to implement and improve a Graph Neural Network (GNN). The chosen model "Anomal-E: A self-supervised network intrusion detection system based on graph neural networks" [18] will be evaluated alongside with data pre-processing to visualize the performance in NIDS.

1) Improving Anomal-E through Enhanced Pre-processing:

a) Insights and Ideas: In the second phase of our project, we aim to improve the effectiveness of the Anomal-E model in NIDS by focusing on pre-processing techniques, while keeping the deep learning components intact. We observed that the original approach lacked sufficient attention to data pre-processing, mainly relying on downsampling the dataset to 10% without further feature fusion, selection, or data balancing. Our goal is to investigate whether these pre-processing strategies can enhance anomaly detection accuracy. The emphasis on pre-processing stemmed from our observation that the original model suffered from a few key issues: low recall rates for detecting attacks (around 90%), which could indicate that the model wasn't exposed enough to attack scenarios, and potential imbalances in the dataset. This imbalance likely contributed to the low recall rate for label 1 (attack samples) and better recall for normal traffic (label 0). Our goal was to address these issues through various pre-processing techniques.

We improved the GitHub codebase's readability and added detailed comments. Challenges included setting up a compatible Python environment with necessary libraries, ensuring CUDA support for GPU computation, and avoiding compatibility issues with library versions.

2) Methods for Improvement:

a) Data Balancing: To address the imbalance in the NF-CSE-CIC-IDS2018-v2 dataset, we employed Random OverSampling (ROS) technique. The objective was to achieve an equal distribution of attack and benign samples while preserving the proportional representation of various attack types. This strategy aimed to minimize bias during training and enhance the model's generalization.

Random OverSampling (ROS) The Random OverSampling method involves oversampling the minority class, the attack samples (samples with label 1), such that the total number of attack samples matches the number of benign samples.

Algorithm 3 Data Balancing with ROS

Require: X_{train}: Training features, y_{train}: Training labels
Ensure: X_{balanced}, y_{balanced}: Balanced features and labels
1: Combine X_{train} and y_{train} into a single DataFrame
2: Separate benign and attack samples
3: Determine the total number of benign samples
4: Determine target distribution from original proportions.
5: Configure the oversampling strategy using calculated targets
6: Apply Random OverSampler to oversample the attack samples
7: Reconstruct the balanced dataset
8: Shuffle the balanced dataset
9: Return the balanced features and labels

Random overSampler ensures balanced training data by maintaining consistent proportional distribution of each attack type post-oversampling, mitigating class imbalance.

b) Feature Selection: Feature selection aims to reduce the complexity of the dataset, potentially lowering training time and improving model performance. We used XGBoost, an algorithm from the first part of the project III-C, adapted for Anomal-E's pre-processing.

3) Contamination and Critique: The original paper referenced a contamination rate of 0.04 for the NF-UNSW-NB15-v2 and NF-CSE-CIC-IDS2018-v2 datasets. However, the NF-CSE-CIC-IDS2018-v2 dataset has a natural contamination rate of 11.95%, raising concerns about adjusting it to 4%. This inconsistency suggests that comparisons using a 0.04 contamination rate might be unreliable. Our experiments showed that contamination could degrade performance, suggesting that a training set with only benign data might work better for the Anomal-E model. To avoid additional discrepancies, we focus on experiments with a 0% contamination rate. Future studies should ensure contamination rates align with the natural dataset distribution for consistency.

4) Exclusion of CBLOF: Among the four classification methods (CBLOF, HBOS, PCA and IF) provided in the original paper, we excluded CBLOF was excluded in our experiments due

to persistent errors related to cluster separation, potentially due to extensive downsampling.

IV. Experiments and Results

A. Feature Fusion

In our experiment, we used the full 42-feature space of UNSW-NB15 for multiclass configuration. Then, we applied XGBoost for feature selection, producing a 19-feature vector 1. We employed Decision Tree for both processes, with XGBoost classifier using default scikit-learn library and parameters..

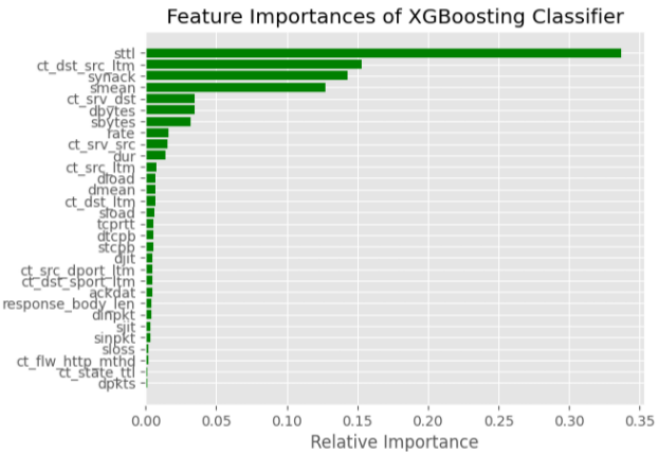


Fig. 1. Important feature selection

TABLE I
Results using 42 features

Metrics	Value		
	Train	Validation	Test
Accuracy	0.9384	0.8524	0.7351
Precision	0.9410	0.8544	0.7197
Recall	0.9384	0.8524	0.7351
F1-score	0.9357	0.8521	0.7081

TABLE II
Results using 19 features

Metrics	Value		
	Train	Validation	Test
Accuracy	0.9376	0.8586	0.7431
Precision	0.9415	0.8608	0.7396
Recall	0.9376	0.8586	0.7431
F1-score	0.9352	0.8579	0.7193

If we see Tables I and II, performance has improved for feature selection method. Though the performance didn't increase drastically, it has reduce the number of features to be trained on, so the training time would be improved greatly.

B. Feature Fusion

This section evaluates our proposed deep learning models and highlights the effectiveness of our data fusion approach compared to traditional methods. We employed the Keras tuner package to optimize hyperparameters, and applied L2 regularization (at 0.01) across all models. Each model underwent training for 10 epochs with a batch size of 32.

TABLE III
with traditional network

Metrics	Value		
	Train	Validation	Test
Loss	0.9240	1.2302	2.3623
Accuracy	0.7176	0.6809	0.3193
Precision	0.9522	0.9289	0.7540
Recall	0.7615	0.6965	0.2398

TABLE IV
with feature fusion mechanism

Metrics	Value		
	Train	Validation	Test
Loss	0.5329	0.5567	0.7214
Accuracy	0.8387	0.8282	0.7680
Precision	0.9897	0.9882	0.8990
Recall	0.9375	0.9307	0.8834

If we see Tables III and IV, performance has greatly improved for feature fusion method. By this experiment, it can be proved that, when the models are trained, we should consider the relationship between different data types of the features and train the models accordingly.

C. Anomal-E

We ran the baseline experiments using only 1% of the dataset due to computational limitations.

The deep learning model (e-GraphSAGE and DGI) was trained with consistent hyperparameters as the original study: a single layer with 256 hidden units, a learning rate of $1e^{-3}$, ReLU activation

TABLE V
PRECISION AND RECALL FOR VARIOUS ANOMALY DETECTION ALGORITHMS

Approach	Algorithm	Precision		Recall	
		Label 0	Label 1	Label 0	Label 1
Baseline	HBOS	0.9803	0.9904	0.9989	0.8520
	PCA	0.9775	0.9901	0.9989	0.8310
	IF	0.9772	0.9893	0.9988	0.8288
XGBoost	HBOS	0.9769	0.9894	0.9988	0.8266
	PCA	0.9769	0.9895	0.9988	0.8266
	IF	0.9769	0.9909	0.9990	0.8266
Data balancing	HBOS	0.9770	0.9900	0.9989	0.8267
	PCA	0.9776	0.9895	0.9988	0.8267
	IF	0.9845	0.9207	0.9896	0.8854

function, Binary Cross-Entropy (BCE) loss function, and Adam optimizer.

Similarly, for the grid search used in the anomaly detection algorithms, we applied the same hyperparameters as the original authors. These included the number of components for PCA (5, 10, 15, 20, 25, 30), the number of estimators for Isolation Forest (20, 50, 100, 150), and the number of bins for HBOS (5, 10, 15, 20, 25, 30).

a) Feature Selection with XGBoost: Initially, we anticipated that reducing features from 42 to 29 using XGBoost would decrease model training time. However, the reduction from 13 minutes and 4 seconds to 12 minutes and 53 seconds was not significant. This suggests that feature selection may not greatly impact training time with a smaller dataset portion, but could be more impactful with a larger dataset. While XGBoost achieved comparable results to the baseline, its overall performance showed slightly lower recall and precision. This indicates that feature reduction did not lead to significant improvements in this scenario.

b) Data Balancing and Model Performance: Regarding data balancing, the recall for label 1 with IF was 0.8854, higher than the best recall obtained by the baseline, which was 0.8520 with HBOS. This improvement in recall suggests that data balancing can help the model detect more anomalies since it has been exposed to more attack samples during the training.

Data balancing did not degrade overall precision, even though the baseline generally had slightly better results. The lowest precision was 0.9207 with IF, a value consistent with the higher recall achieved with the same algorithm (0.8854). This aligns with the typical tradeoff between recall and precision in anomaly detection: higher recall captures more anomalies but may increase false positives, affecting precision, while higher precision reduces false positives but may miss some anomalies due to lower recall.

Balancing recall and precision is vital in network anomaly detection to detect most threats while minimizing false alarms, ensuring effective security systems with manageable false positives.

V. CONTRIBUTION AND DISCUSSION

Our study enhances network intrusion detection research by:

- We show that strategic feature engineering can improve ML model accuracy
- Our study indicates that feature fusion significantly boosts DL model performance in detecting complex threats, suggesting its vital role in model development.
- We applied different data pre-processing techniques to see its affect the performance of the GNN model-based approach in NIDS that we chose to study.

These insights underscore the importance of advanced feature techniques and the careful selection of models to effectively counter cyber threats.

VI. CONCLUSION

Our investigation into feature selection, fusion, and model comparison highlights their critical importance in network intrusion detection. Through evaluating ML and GNN models, we demonstrate the potential of advanced data processing and novel architectures in improving detection capabilities against complex threats. These insights also underscore the need for sophisticated feature manipulation and different approaches.

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